

Chronic Interstitial Nephritis

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DEFINITION

Chronic interstitial nephritis is a histologic entity characterized by progressive scarring of the tubulointerstitium, with tubular atrophy, macrophage and lymphocytic infiltration, and interstitial fibrosis. Because the degree of tubular damage accompanying interstitial nephritis is variable, the term *tubulointerstitial nephritis* is used interchangeably with *interstitial nephritis*. *Tubulitis* refers to infiltration of the tubular epithelium by leukocytes, usually lymphocytes.

There are many primary and secondary causes of chronic interstitial nephritis (Box 65.1). Tubulointerstitial injury is clinically important because it is a better predictor than the degree of glomerular injury of present and future renal function. Although any glomerular disease can cause secondary injury to the tubulointerstitium through the effects of proteinuria and ischemia, in this chapter we discuss only primary chronic interstitial nephritis.

PATHOGENESIS

The tubulointerstitium can be injured by toxins (e.g., heavy metals), drugs (e.g., analgesics), crystals (e.g., calcium phosphate, uric acid), infections, obstruction, immunologic mechanisms, and ischemia. Regardless of the initiating mechanism, the tubulointerstitial response shows little variation. Tubular injury results in the release of chemotactic substances and the expression of leukocyte adhesion molecules that attract inflammatory cells into the interstitium. Tubular cells express human leukocyte antigens, serve as antigen-presenting cells, and secrete complement components and vasoactive mediators, all of which may further stimulate or attract macrophages and T cells. Growth factors released by tubular cells and macrophages, such as platelet-derived growth factor and transforming growth factor β , may stimulate fibroblast proliferation and activation, leading to matrix accumulation.¹ The source of fibroblasts in renal interstitial fibrosis remains controversial but may include an intrinsic fibroblast (perivascular Gli1-positive progenitors) population, migration of circulating fibrocytes from perivascular areas, and phenotypic transition of pericytes into fibroblasts.² Over time, a loss of peritubular capillaries and decreased oxygen diffusion caused by expansion of the interstitium render the kidney hypoxic, and progressive apoptosis leads to local hypocellularity and fibrosis.² Kidney function can become severely decreased, and kidney replacement therapy may be required.

EPIDEMIOLOGY

Whereas chronic interstitial nephritis occurs with progressive kidney disease of all causes, primary chronic interstitial nephritis is not a common cause of end-stage kidney disease (ESKD); reports range from 42% in Scotland to 3% to 4% in China and the United States.^{3–5} This

variability in reported incidence may relate to differences in how diagnoses are made, exposure to causal factors, and treatment modalities, such as the choice of antibiotics and the indication for pain killers.

Marked increases in the incidence of chronic interstitial nephritis have been reported in certain areas. Chronic kidney disease of unknown etiology/uncertain cause (CKDu) is a type of CKD that mainly affects agricultural communities in specific areas of the world. These regional nephropathies include Sri Lanka and India, some coastal areas of Central America, and the Balkan region in Europe (see Chapters 66 and 79).

PATHOLOGY

The pathologic features of chronic interstitial nephritis are nonspecific. They include tubular cell atrophy or dilation; interstitial fibrosis that is composed of interstitial (types I and II) collagens; and mononuclear cell infiltration with macrophages, T cells, and occasionally other cell types (neutrophils, eosinophils, and plasma cells). Tubular lumina vary in diameter but may show marked dilation, with homogeneous casts producing a thyroid-like appearance, hence the term *thyroidization*.

Some forms of chronic interstitial nephritis are associated with granulomatous lesions. Whereas a noncaseating granulomatous pattern is typical of sarcoidosis, interstitial granulomatous reactions also occur in response to infection of the kidney by mycobacteria (Fig. 65.1), fungi, or bacteria; certain drugs (including rifampin, sulfonamides, proton pump inhibitors [PPIs], and narcotics); and oxalate or uric acid crystal deposition.⁶ Interstitial granulomatous reactions have been noted in renal malacoplakia (Chapter 53), granulomatous polyangiitis, heroin abuse, and after jejunoileal bypass surgery. The etiology for granulomatous interstitial nephritis remains obscure in 10% of cases.

CLINICAL MANIFESTATIONS

The impaired kidney function is often insidious, and the early manifestations of the disease are those of tubular dysfunction, which may go undetected (Box 65.2).⁷ Diagnosis is often made incidentally on routine laboratory screening or during evaluation of hypertension, in association with reduced glomerular filtration rate (GFR). Proteinuria is commonly less than 1 g/day. Urinalysis may show only occasional white blood cells (WBCs) and, rarely, WBC casts. Hematuria is uncommon. Anemia may occur relatively early because of dysfunction or loss of renal erythropoietin-producing cells in the interstitium.

The tubular dysfunction is often generalized, but some conditions may manifest with proximal tubular defects, including aminoaciduria, phosphaturia, proximal renal tubular acidosis (RTA), or, rarely, complete Fanconi syndrome (see Chapter 50). Distal tubular defects can be associated with distal RTA (type 1 or type 4; see Chapter 13).

BOX 65.1 Major Causes of Chronic Interstitial Nephritis^a

Diseases in Which the Kidneys Are Macroscopically Normal

- Drugs and toxins (e.g., aristolochic acid, lithium, cyclosporine, tacrolimus, indinavir, cisplatin, proton pump inhibitors)
- Metabolic (hyperuricemia, hypokalemia, hypercalcemia, hyperoxaluria, cystinosis)
- Heavy metals (lead, cadmium, arsenic, mercury, gold, uranium)
- Radiation
- Balkan nephropathy
- Mesoamerican nephropathy
- Immune-mediated conditions (systemic lupus erythematosus, Sjögren syndrome, sarcoidosis, granulomatous polyangiitis, other vasculitides)
- Vascular diseases (atherosclerotic kidney disease) (see [Chapter 43](#))
 - Transplantation (chronic transplant rejection)
 - Hematologic disturbances (multiple myeloma, light-chain deposition disease, lymphoma, sickle cell disease, paroxysmal nocturnal hemoglobinuria) (see [Chapters 51 and 68](#))
- Progressive glomerular disease of any cause (glomerulonephritides, diabetes, hypertension)
- Idiopathic

Diseases in Which the Kidneys Are Macroscopically Abnormal

- Analgesic nephropathy
- Chronic obstruction (see [Chapter 61](#))
- Hereditary (nephronophthisis, medullary cystic kidney disease, familial juvenile hyperuricemic nephropathy, autosomal dominant polycystic kidney disease, autosomal recessive polycystic kidney disease)
- Infection (chronic pyelonephritis, malacoplakia, xanthogranulomatous pyelonephritis; see [Chapter 53](#))

^aKidneys of any clinical entity can be shrunken at end-stage. Some diseases categorized as macroscopically normal can, in later stages, result in macroscopically abnormal kidneys. For example, kidneys of sickle cell nephropathy are macroscopically normal unless papillary necrosis is present.

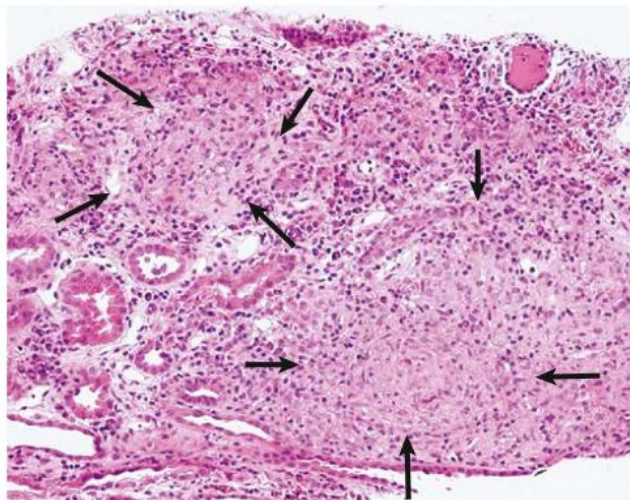


Fig. 65.1 Renal Tuberculosis. Noncaseating granulomas with epithelioid cells in miliary tuberculosis (arrows show the peripheries of the granulomas). Although the typical pathologic change is granuloma with caseous necrosis with Langerhans-type giant cells, these atypical granulomas can be observed in tuberculosis and should be differentiated from sarcoidosis. (H&E stain.) (Courtesy Dr. Noriko Uesugi, Ibaraki, Japan.)

BOX 65.2 Functional Manifestations of Chronic Interstitial Nephritis

- Deterioration of glomerular filtration rate with insidious onset
- Tubular proteinuria mainly composed of low-molecular-weight protein (generally <1 g/day)
- Inactive urinary sediment
- Anemia of chronic kidney disease at a relatively early stage
- Proximal tubular dysfunction (aminoaciduria, phosphaturia, proximal renal tubular acidosis, Fanconi syndrome)
- Distal tubular dysfunction (type IV renal tubular acidosis)
- Medullary dysfunction (concentrating defects)
- Salt-wasting syndrome
- Salt-sensitive hypertension

Concentrating defects (increased urinary frequency and nocturia) can be a sign of medullary dysfunction and may be severe enough to result in nephrogenic diabetes insipidus. Some patients develop a salt-wasting syndrome. Others, particularly with microvascular disease, may have a relative inability to excrete salt with resultant salt-sensitive hypertension.⁸

Clues to the causes of tubulointerstitial nephritis by history and physical examination are shown in [Table 65.1](#).⁹

TREATMENT

Treatment includes identification and elimination of any exogenous agents (drugs, heavy metals), metabolic causes (hypercalcemia), or conditions (obstruction, infection) potentially causing the chronic interstitial lesion. In addition to classic drugs and toxins, a growing list of medications (e.g., PPIs) is now being linked to the risk for developing CKD, which emphasizes the need to consider the risk and benefit of any drug carefully, particularly in patients with a progressive course. Specific treatments may be required for an underlying cause of chronic interstitial nephritis, such as corticosteroids for sarcoidosis. General measures include control of blood pressure. Most clinicians favor the use of angiotensin-converting enzyme (ACE) inhibitors or angiotensin receptor blockers (ARBs), which reduce glomerular and systemic pressures, decrease proteinuria, and increase renal blood flow. However, the role of ACE inhibitors and ARBs is less clear when blood pressure is normal or when there are excessive solute losses (such as salt-wasting). Specific therapies for each clinical entity are discussed later.

DRUG-INDUCED CHRONIC INTERSTITIAL NEPHRITIS

Several drugs and herbs can cause chronic interstitial nephritis. Cyclosporine- and tacrolimus-induced nephropathy are discussed in [Chapter 106](#); aristolochic acid as a cause of aristolochic acid–associated nephropathy (formerly known as *Chinese herbs nephropathy* and *Balkan nephropathy*) is discussed in [Chapter 79](#).

Lithium Nephropathy

Definition and Epidemiology

Lithium is commonly used in the treatment of bipolar disorder. Complications of lithium treatment include nephrogenic diabetes insipidus, acute lithium intoxication, and chronic lithium nephrotoxicity. A meta-analysis of the data of 14 studies involving 1172 patients receiving chronic lithium therapy showed that the prevalence of reduced GFR was 15%.¹⁰

TABLE 65.1 Clues to Causes of Tubulointerstitial Nephritis by History and Physical Examination

Feature	Symptom, Sign, or Historical Clue	Potential Diagnosis
Occupational history	Exposure to heavy metals (e.g., batteries, alloys)	Lead or cadmium nephropathy
Alcohol	History of moonshine ingestion	Lead nephropathy
Social history	Country of origin	Balkan nephropathy
Past history	Systemic lupus erythematosus	Disease-associated chronic interstitial nephritis
	Sjögren syndrome	
	Sarcoidosis	
	Inflammatory bowel disease	
	Autoimmune pancreatitis	
	Chronic pain syndrome	Analgesic nephropathy
	Gouty attack	Lead nephropathy
Medication	Prescribed	Drug-induced chronic interstitial nephritis
	OTC (NSAIDs, PPIs)	Analgesic
	Herbal	Aristolochic acid-associated nephropathy
	Indinavir	Crystal nephropathy
Physical examination	Dry eyes	Sjögren syndrome
	Uveitis	TINU syndrome
Laboratory examination	Hyperuricemia	Chronic uric acid nephropathy
	Hypokalemia	Hypokalemic nephropathy
	Hypercalcemia	Hypercalcemic nephropathy
	High serum IgG4 levels	IgG4-related sclerosing disease
Radiologic examination	Decreased volume, bumpy contours, and papillary calcification on CT	Analgesic nephropathy
	Microcysts on MRI or ultrasound	Lithium nephropathy
	Nephrocalcinosis on CT	Hypercalcemic nephropathy

CT, Computed tomography; IgG, immunoglobulin G; MRI, magnetic resonance imaging; NSAIDs, nonsteroidal antiinflammatory drugs; OTC, over the counter; PPIs, proton pump inhibitors; TINU, tubulointerstitial nephritis and uveitis.

Modified from Beck LH Jr, Salant DJ. Glomerular and tubulointerstitial diseases. *Prim Care*. 2008;35:265–296.

Pathogenesis

Diabetes insipidus results from accumulation of lithium in the collecting tubular cells after entry into these cells through sodium channels in the luminal membrane. Lithium blocks vasopressin-induced reabsorption by inhibiting adenylate cyclase activity, and hence cyclic adenosine monophosphate production, and also by decreasing the apical membrane expression of aquaporin 2, the collecting tubule water channel. Chronic lithium-induced interstitial nephritis may occur, possibly because of inositol depletion and inhibition of cell proliferation.

Pathology

Biopsies show focal chronic interstitial nephritis with interstitial fibrosis, tubular atrophy, and glomerulosclerosis. Although similar histologic changes have been reported in psychiatric patients without a history of lithium therapy, patients with lithium exposure often show microcystic changes in the distal tubule; interstitial inflammation and vascular changes are relatively minor. The degree of interstitial fibrosis is related to the duration of administration and cumulative dose.

Clinical Manifestations

Lithium-associated diabetes insipidus. The most common presentation of lithium-induced nephrotoxicity is nephrogenic diabetes insipidus, characterized by resistance to vasopressin, polyuria, and polydipsia. Impaired urinary concentrating ability is found in about 50% of patients, and polyuria resulting from nephrogenic diabetes insipidus occurs in about 20% of patients chronically treated with lithium.

Lithium is also rarely a cause of hypercalcemia, which could potentially exaggerate the tubular concentrating defect and contribute to

the development of chronic interstitial nephritis in lithium-treated patients. Nephrogenic diabetes insipidus in lithium treatment may be associated with distal RTA, although this partial functional defect is virtually never of clinical importance.

Chronic lithium nephropathy. In a retrospective analysis of data from the United Kingdom, the presence of lithium in serum was associated with an increased risk for stage 3 CKD (hazard ratio 1.93), with females aged younger than 60 years and those with lithium concentrations higher than median at increased risk.¹¹ Nephrogenic diabetes insipidus induced by lithium may persist despite the cessation of treatment, indicating irreversible renal damage.

In one study, the mean serum creatinine concentration of patients with biopsy-proven chronic lithium nephrotoxicity was 2.8 mg/dL (247 μmol/L) at the time of biopsy, and 42% of patients had proteinuria greater than 1 g/day.¹² After kidney biopsy, all but one patient discontinued treatment with lithium, but seven patients nevertheless progressed to ESKD. A study of 74 lithium-treated patients in France showed that lithium-induced nephropathy developed slowly over several decades, with an average latency between the start of therapy and ESKD of 20 years.¹¹

Magnetic resonance imaging, in particular the half-Fourier acquisition single-short turbo spin-echo T2-weighted sequence, without the use of gadolinium, or ultrasound may help in detection of the characteristic microcysts in the kidney.¹³

Treatment

After other potential causes of polyuria and polydipsia have been excluded, particularly psychogenic polydipsia, the first step to consider

is a reduction in lithium dosage. The potassium-sparing diuretic amiloride improves the polyuria and also blocks lithium uptake through sodium channels in the collecting duct. Thiazide diuretics should be avoided because they increase the risk for acute lithium intoxication because of the resultant volume contraction and an increase in sodium and lithium reabsorption in the proximal tubule. ACE inhibitors are generally not used because of the risk for hypotension and worsening GFR in the setting of volume depletion.

Patients undergoing long-term lithium treatment should have kidney function (serum creatinine and estimated GFR) and 24-hour urine volume measured yearly. Lithium has a narrow therapeutic index, so levels should be monitored and maintained between 0.6 and 1.25 mmol/L. The severity of chronic lithium intoxication correlates directly with the serum lithium concentration and may be categorized as mild (1.5–2.0 mmol/L), moderate (2.0–2.5 mmol/L), or severe (>2.5 mmol/L). Once-daily regimens are less toxic than multiple daily administration schedules, perhaps because of the possibility of renal tubular regeneration with a once-daily administration schedule.¹⁴ Prevention of volume depletion is also important.

Because progressive kidney injury with reduced GFR in patients without prior acute lithium intoxication is relatively unusual, reductions in GFR initially should be treated by decreasing the dose of lithium. If GFR is persistently reduced, a kidney biopsy should be considered, although the findings rarely mandate the cessation of lithium treatment. At all times, the risk for discontinuation in a patient with a severe unipolar or bipolar affective disorder needs to be balanced against the relatively low risk for progressive kidney injury.

Analgesic Nephropathy

Definition and Epidemiology

Analgesic nephropathy resulted from the abuse of analgesics, commonly mixtures containing phenacetin, aspirin, and caffeine, that were available as over-the-counter preparations in Europe and Australia. It is now extremely rare; indeed, some doubt that new cases are still presenting, after restrictions in compound analgesic sales.¹⁵ A large study in the United States showed no association between use of current analgesic preparations and increased risk for chronic kidney dysfunction,¹⁶ although it does increase the risk for acute kidney injury (see Chapter 70).

Pathogenesis and Pathology

The primary injury in analgesic nephropathy is medullary ischemia caused by toxic concentrations of phenacetin metabolites combined with relative medullary hypoxia, aggravated by inhibition of vasodilatory prostaglandin synthesis and glutathione (an antioxidant) levels. The main pathologic consequence is papillary necrosis, with secondary tubular atrophy, interstitial fibrosis, and a mononuclear cellular infiltrate (Fig. 65.2).

Clinical Manifestations

Analgesic nephropathy is five to seven times more common in females than in males. Kidney manifestations are nonspecific and consist of slowly progressive CKD with impaired urine concentrating ability, urinary acidification defects, and impaired sodium conservation. Patients may present with hypertension, reflecting the use of nonsteroidal anti-inflammatory drugs (NSAIDs), and degree of kidney parenchymal injury. Urinalysis shows sterile pyuria and mild proteinuria. Patients with analgesic nephropathy are at increased risk for transitional cell carcinoma of the uroepithelium. A recent prospective analysis of the Nurses' Health Study and the Health Professionals Follow-up Study also showed an association of regular use of nonaspirin NSAIDs with an increased risk of renal cell carcinoma, but aspirin and acetaminophen use were not implicated.¹⁷

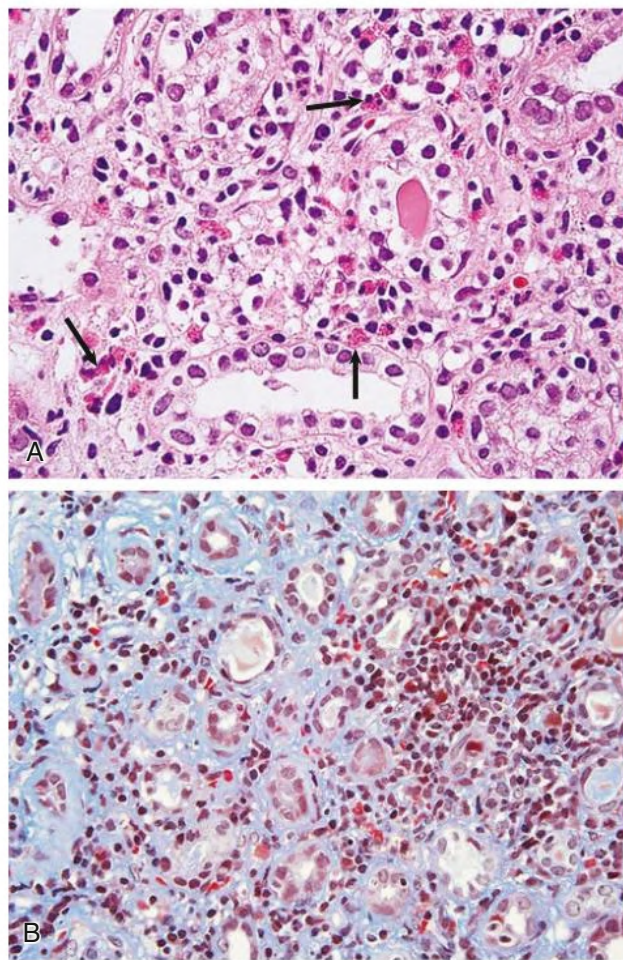


Fig. 65.2 Histologic Changes in Analgesic Nephropathy. (A) Interstitial nephritis in a patient with analgesic nephropathy associated with marked mononuclear cellular infiltrate including eosinophils (arrows). (H&E stain; original magnification $\times 600$.) (B) Analgesic nephropathy with interstitial fibrosis and inflammatory cell infiltration. (Masson trichrome stain; original magnification $\times 400$.) (Courtesy Drs. Akira Shimizu and Hideki Takano, Nippon Medical School, Tokyo.)

Diagnosis

Papillary necrosis is present histologically in almost all patients, but it can be detected radiologically only if part or all of the papilla has sloughed. Papillary necrosis is not pathognomonic of analgesic nephropathy; it is also seen in diabetic nephropathy (particularly during an episode of acute pyelonephritis), sickle cell nephropathy, urinary tract obstruction, and renal tuberculosis. Non-contrast-enhanced computed tomography (CT) demonstrates a decrease in renal mass with either bumpy contours or papillary calcifications (Fig. 65.3; see also Chapter 51).¹⁸

Treatment

Management consists of stopping or at least reducing the intake of analgesic medications. Because of the increased incidence of uroepithelial tumors, close follow-up, including regular urinalysis, is necessary. New hematuria requires early referral for urologic evaluation, which includes urinary cytology, cystoscopy, and CT scan. It would be prudent to perform yearly urinary cytologic examination at least several years after cessation of analgesics.

Analgesic nephropathy associated with over-the-counter medicines is also discussed in Chapter 79.

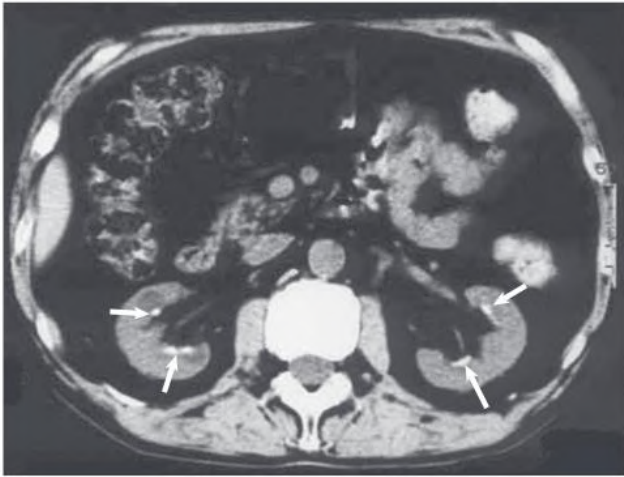


Fig. 65.3 Papillary Calcifications in Analgesic Nephropathy. Non-contrast-enhanced computed tomography scan of a patient with long-time analgesic abuse shows thinning of the renal parenchyma and typical papillary calcifications (arrows). (Courtesy Dr. Yoshifumi Ubara, Toranomon Hospital, Tokyo.)

CHRONIC INTERSTITIAL NEPHRITIS CAUSED BY METABOLIC DISORDERS

Metabolic disorders that cause interstitial nephritis are discussed here. Hyperoxaluria is described in [Chapter 60](#) and cystinosis in [Chapter 50](#).

Chronic Uric Acid Nephropathy

Definition and Epidemiology

Nearly half of patients with gout have reduced GFR, and most have evidence for CKD at autopsy. An elevated serum uric acid has also been consistently shown to predict the development of CKD.¹⁹ This has led many investigators to propose that hyperuricemia and/or gout may be a cause of CKD,²⁰ either from the deposition of urate crystals in the kidney or from effects of soluble uric acid to active inflammatory processes. Nevertheless, two recent clinical trials of urate-lowering therapy in subjects with CKD were negative, challenging whether chronic uric acid nephropathy exists.^{21,22}

Pathogenesis

Chronic gout and severe hyperuricemia can be associated with the deposition of uric acid crystals in the renal medulla ([Fig. 65.4](#)). A recent study suggests this may be present in as many as a third of subjects with gout and that it can be diagnosed by ultrasound findings of a hyperechoic renal medulla.²³ The deposition of urate crystals can cause injury to the tubules and rupture into the interstitium where they cause inflammasome activation with inflammation and fibrosis. Experimental studies also suggest that high levels of soluble uric acid may also cause chronic kidney injury by activating the renin-angiotensin system and inducing oxidative stress, resulting in microvascular disease, glomerular hypertension, and impaired renal autoregulation.²⁴

Pathology

Renal functional abnormalities are observed in 30% to 50% of patients who have had gout for many years, and histologic changes are observed in more than 90%.²⁵ The most consistent histologic findings are arteriosclerosis, focal or global glomerulosclerosis, and chronic tubulointerstitial disease. Uric acid crystals also can be occasionally found within tubules and in the interstitium (see [Fig. 65.4](#)), especially in the outer

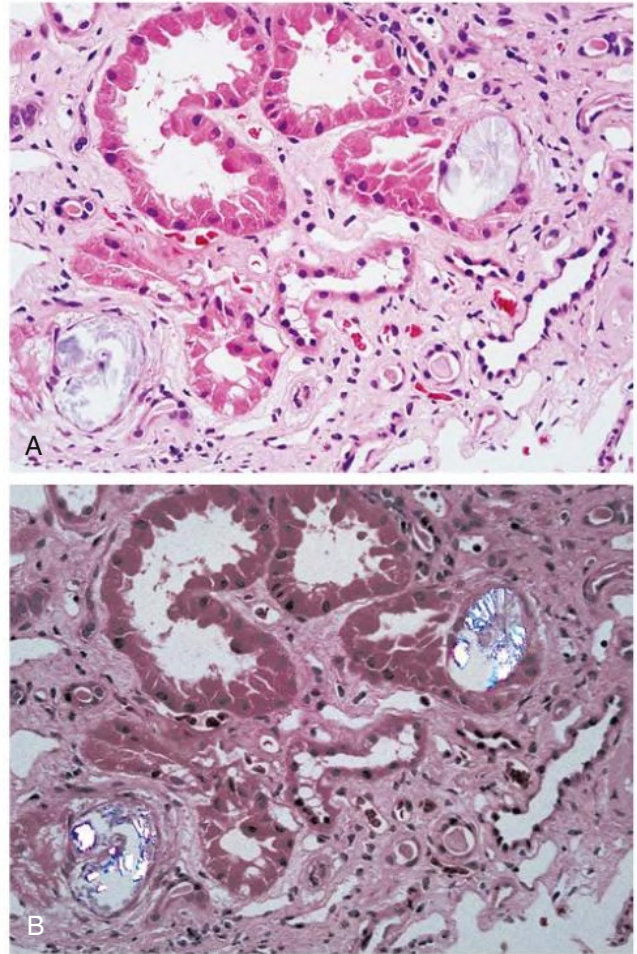


Fig. 65.4 Chronic Uric Acid Nephropathy. (A) Large collections of elongated or fragmented uric acid crystals are present in association with atrophic tubules. (H&E stain; original magnification $\times 400$.) (B) The crystalline masses are refractile under polarized light. (Courtesy Drs. Akira Shimizu and Hideki Takano, Nippon Medical School, Tokyo.)

medulla, and, on rare occasions, medullary kidney tophi can be found on gross anatomic dissection.

Clinical Manifestations

Patients with gout often present with hypertension with mildly impaired kidney function, mild proteinuria, unremarkable urinary sediment, and minor tubular dysfunction (usually impairment of urine-concentrating ability manifested as isosthenuria). Approximately one-third of subjects with CKD also have a history of gout, and nearly 50% have hyperuricemia. A role for uric acid in CKD should particularly be considered if there is a disproportionate elevation in serum uric acid in relation to the degree of GFR reduction ([Table 65.2](#)).²⁶

Diagnosis

The most important differential diagnosis for chronic uric acid nephropathy is chronic lead nephropathy. Familial juvenile hyperuricemic nephropathy is a rare autosomal dominant disease that mimics chronic gouty nephropathy but manifests in adolescence or during early childhood (see [Chapter 50](#)).

Treatment

It remains controversial whether lowering uric acid can improve kidney disease in patients with gout or hyperuricemia. One randomized trial

TABLE 65.2 Serum Creatinine and Uric Acid Levels in Chronic Kidney Disease

SERUM CREATININE		SERUM URIC ACID	
mg/dL	μmol/L	mg/dL	μmol/L
<1.5	<132	9	536
1.5–2.0	132–176	10	595
>2.0	>176	12	714

Serum uric acid levels are increased with decreased glomerular filtration rate, but if uric acid levels are especially high for the level of creatinine, then chronic urate nephropathy should be considered. Shown are threshold levels of uric acid for which higher levels should increase the suspicion for chronic urate nephropathy.

Modified from Badve SV, Pascoe EM, Tikun A, et al.; CKD-FIX Study Investigators. Effects of allopurinol on the progression of chronic kidney disease. *N Engl J Med*. 2020;382:2504–2513.

demonstrated that allopurinol therapy is associated with preservation of eGFR in CKD, although the treatment did not show any effects on the defined study endpoint, which was ESKD.²⁷ Withdrawal of allopurinol from patients with stable CKD has resulted in worsening of hypertension and acceleration of kidney dysfunction, especially in patients who were not on ACE inhibitors.²⁸ Lowering uric acid in patients with asymptomatic hyperuricemia is also associated with an increase in eGFR.²⁹ There also have been reports that lowering uric acid may reduce the risk for cardiovascular disease in patients with CKD.^{27,30} Recently, two placebo-controlled randomized trials did not find lowering uric acid to be of benefit in subjects with type 1 diabetes and CKD (the PERL study)²¹ and in subjects with CKD (CKD-Fix Trial).²² However, both studies recruited subjects with normal uric acid levels in which benefit was not expected, and both excluded subjects with a history of gout, which was the group most expected to benefit.

We still recommend lowering uric acid in subjects with CKD who have markedly elevated serum uric acid levels (>8–9 mg/dL) or a history of gout with evidence of active progression of CKD. To reduce the risk for a hypersensitivity reaction (Stevens-Johnson–like syndrome), allopurinol should be initiated at a dose of 50 to 100 mg/day, increasing to 200 or 300 mg/day after several weeks as tolerated. The allopurinol hypersensitivity syndrome is more common in subjects who carry HLA-**B58:01*, and typing subjects before initiation of treatment may be of benefit in subjects who are Han Chinese or Korean that are more likely to carry this HLA antigen.

The newer xanthine oxidase inhibitor febuxostat does not require modification of dose at lower eGFR and appears to be less frequently associated with hypersensitivity or nephrotoxicity, but there are some studies that suggest that its use may be associated with greater cardiovascular mortality than that observed with allopurinol.³¹

Hypokalemic Nephropathy

Definition and Epidemiology

Hypokalemia, if persistent for prolonged periods, can induce kidney cysts, chronic interstitial nephritis, and progressive loss of kidney function, known as *hypokalemic nephropathy*, which can be inherited or acquired. Hypokalemic nephropathy occurs in 15% to 20% of individuals with anorexia nervosa.³²

Pathology

The characteristic finding is vacuolation of the kidney tubules as a result of dilation of cisternae of the endoplasmic reticulum and basal folding, which is usually limited to the proximal tubular segments

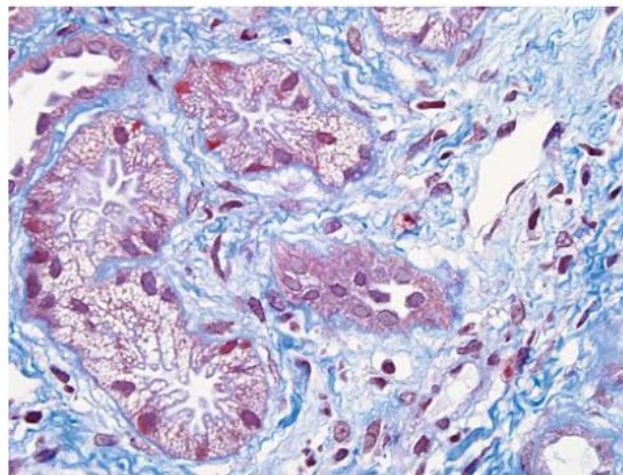


Fig. 65.5 Hypokalemic Nephropathy. Vacuolization of the renal tubules is observed in association with interstitial fibrosis in a patient with hypokalemic nephropathy. (Masson trichrome stain; original magnification $\times 400$.) (Courtesy Drs. Akira Shimizu and Hideki Takano, Nippon Medical School, Tokyo.)

(Fig. 65.5). This abnormality generally requires at least 1 month to develop and is reversible with potassium supplementation. More prolonged hypokalemia can lead to more severe changes, predominantly in the kidney medulla, including interstitial fibrosis, tubular atrophy, and cyst formation. There is experimental evidence that hypokalemic injury may be caused by hypokalemia-induced renal vasoconstriction with ischemia. Local ammonia production stimulated by hypokalemia also may lead to intrarenal complement activation that may contribute to the kidney injury. Furthermore, the associated intracellular acidosis can stimulate cell proliferation, which may account for the occasional development of cysts in hypokalemic patients.

Clinical Manifestations

Impaired urinary concentration, manifesting with nocturia, polyuria, and polydipsia, may occur, particularly when serum potassium concentration is consistently below 3.0 mmol/L for months or years. The average duration of hypokalemia reported in patients with chronic hypokalemic nephropathy is 3.5 to 9 years. The renal defect is associated with decreased collecting tubule responsiveness to vasopressin, possibly because of decreased expression of aquaporin 2.

Diagnosis

Although degenerative changes in proximal tubular cells are a consistent but nonspecific finding in hypokalemic nephropathy, a particularly characteristic finding is vacuolar changes in the proximal tubules (see Fig. 65.5). Similar vacuolization of the convoluted tubules is observed in ethylene glycol poisoning.

Treatment

Hypokalemia usually can be treated with oral potassium supplements. The treatment of hypokalemia is discussed in Chapter 10. Coarse cytoplasmic vacuoles may persist for some time after normalization of serum potassium values.

Hypercalcemic Nephropathy

Definition and Epidemiology

Hypercalcemia can cause transient and reversible kidney vasoconstriction with a decrement in kidney function and chronic interstitial nephritis secondary to tubular cell necrosis and intratubular

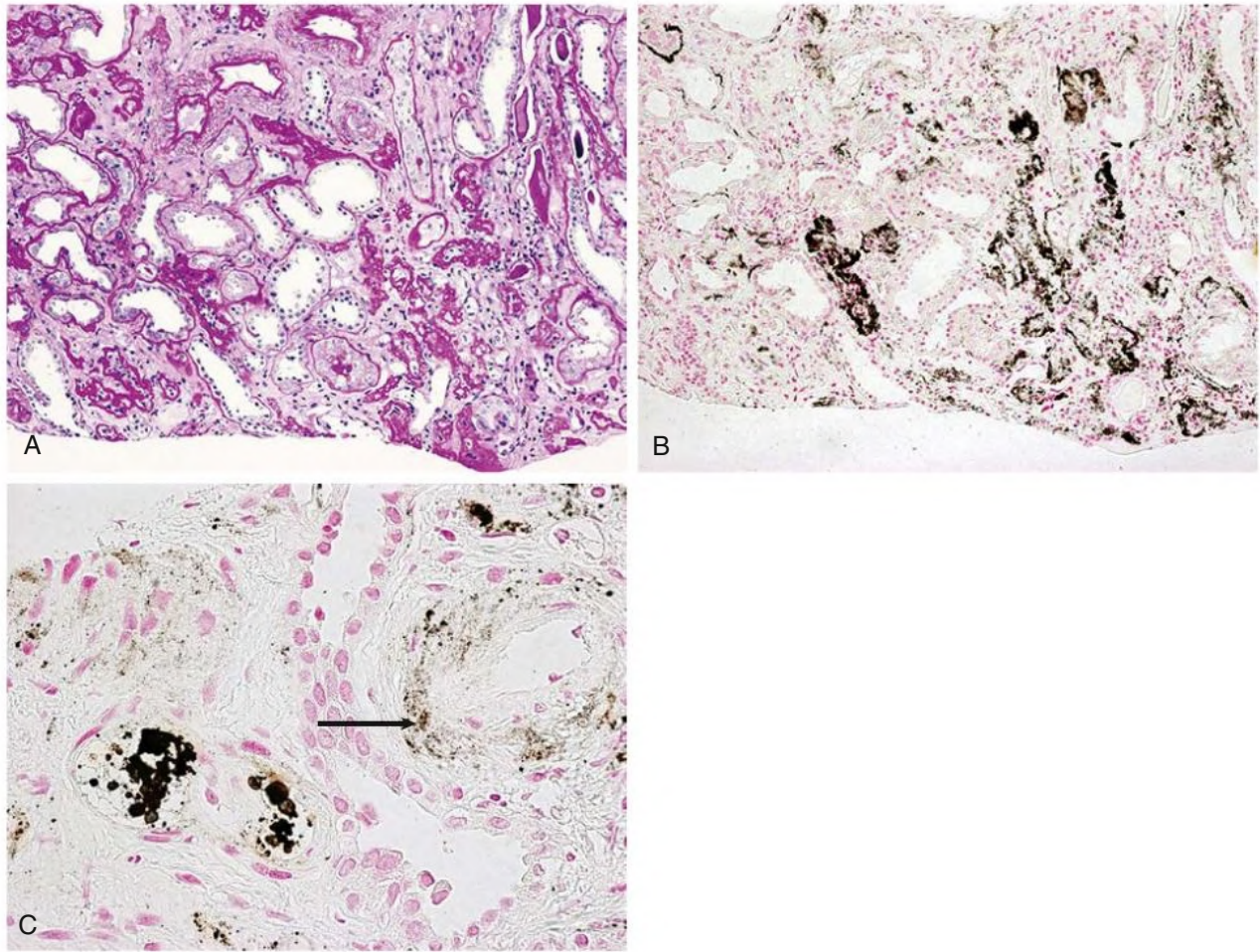


Fig. 65.6 Hypercalcemic Nephropathy Caused by Sarcoidosis. (A) Marked tubular atrophy and interstitial fibrosis with mild lymphocytic infiltrate. (B) Dense calcium deposits are seen in the thickened basement membrane of the atrophic tubules and in the fibrotic area of interstitium (serial section of A). (C) Intraluminal calcium plaque in the atrophic tubules. Granular calcium deposits are observed in the arterial wall (*arrow*). (A, PAS stain. B–C, von Kossa stain.) (Courtesy Dr. Noriko Uesugi, Ibaraki, Japan.)

obstruction. In addition, hypoparathyroidism (especially after surgical treatment of hyperparathyroidism) can result in marked hypercalciuria and a similar syndrome in the absence of hypercalcemia.

Pathology

Focal degeneration and necrosis of the tubular epithelium, primarily in the medulla, where calcium is concentrated, develop soon with persistent hypercalcemia. Although focal degenerative and necrotic lesions of the tubular epithelium can be observed with acute hypercalcemia, the most distinctive histologic feature of long-standing hypercalcemia is calcific deposits in the interstitium (nephrocalcinosis; [Fig. 65.6](#)). Deposition begins in the medullary tubules, followed by deposition in the cortical proximal and distal tubules and within the interstitial space and secondarily leads to mononuclear cell infiltration and tubular necrosis.

Clinical Manifestations

Macroscopic nephrocalcinosis is often detected on radiography or ultrasound. A defect in urinary concentration is the most notable tubular dysfunction and manifests as polyuria and polydipsia. The mechanism is incompletely understood, but the impairment relates both to a reduction in medullary solute content and to interference with the cellular response to vasopressin. Reversible impairment of eGFR can result from either acute or chronic hypercalcemia by decreased renal

blood flow. Irreversible, advanced CKD is a rare consequence of long-standing hypercalcemia and is almost invariably associated with calcium crystal deposition in the interstitium of the kidney.

CHRONIC INTERSTITIAL NEPHRITIS CAUSED BY HEREDITARY DISEASES OF THE KIDNEY

Nephronophthisis (NPHP) and medullary cystic kidney disease (MCKD), or autosomal dominant tubulointerstitial kidney disease (ADTKD) (or the NPHP-MCKD complex) are hereditary diseases associated with renal cysts at the corticomedullary junction. Karyomegalic interstitial nephritis is a hereditary form of chronic tubulointerstitial nephritis histologically characterized by nuclear enlargement with irregular outlines, hyperchromatic aspect, and prominent nucleoli. These disorders are described in detail in [Chapters 47 and 50](#).

CHRONIC INTERSTITIAL NEPHRITIS ASSOCIATED WITH HEAVY METAL EXPOSURE

Lead Nephropathy

Definition and Epidemiology

Acute lead intoxication is rare but may manifest with abdominal pain, encephalopathy, hemolytic anemia, peripheral neuropathy,

and proximal tubular dysfunction (Fanconi syndrome). In contrast, chronic low-level exposure to lead is associated with CKD, often with hyperuricemia. Because lead has a biologic half-life of several decades, both intermittent acute poisoning and low-level environmental exposure result in chronic cumulative lead poisoning.

Although some epidemiologic studies have suggested that low-level exposure may be associated with CKD or hypertension,^{33–35} data indicating that lead causes CKD and ESKD are relatively sparse,³⁶ although there may be regional differences in the observed risk because of geographic variation in exposure. The pathogenesis of the kidney disease may be related to the accumulation of reabsorbed lead in the proximal tubule cells, effects of chronic lead exposure on the vasculature, or lead-induced hyperuricemia. Additional well-controlled longitudinal studies with adequate exposure and effect variables are awaited to confirm that lead exposure causes deterioration of renal function and eventual ESKD.

Pathology

The kidneys are granular and contracted. The characteristic morphology is chronic interstitial nephritis demonstrating nonspecific tubular atrophy, interstitial fibrosis, and a paucity of inflammatory cells. The earliest histologic finding is proximal tubular injury, with intranuclear inclusion bodies composed of a lead-protein complex. Glomerular scarring can be observed as a secondary event, and arteries and arterioles demonstrate medial thickening and luminal narrowing, probably related to hypertension. Immunofluorescence (IF) studies are noncontributory.

Clinical Manifestations

Chronic lead nephropathy is usually identified when a source of high exposure is known (occupational hazard or consumption of illicitly distilled spirits [moonshine]). Hyperuricemia is common because of impaired uric acid excretion. Urine sediment is benign, and urinary protein excretion is less than 2 g/day. Hypertension is almost always present, and in the absence of appropriate testing or a careful exposure history, lead nephropathy is often misdiagnosed as hypertensive kidney disease. Gouty arthritis (“saturnine gout”) affects about half of patients. Patients with chronic lead intoxication may occasionally manifest other signs, including peripheral motor neuropathy, anemia with basophilic stippling, and perivascular cerebellar calcification.

Diagnosis

Lead nephropathy may be underdiagnosed because no simple diagnostic blood test is available. Lead nephropathy is easily confused with chronic uric acid nephropathy, in which uric acid deposits (tophi) may form in the renal interstitium. All patients with hyperuricemia and reduced eGFR should have a history of occupational lead exposure excluded. The blood lead concentration is an insensitive measure of cumulative body stores. A clinical diagnosis of lead nephropathy is based on a history of exposure, evidence of kidney dysfunction, and an abnormal calcium disodium edetate (CaNa_2EDTA) lead chelation test. The association with gout and CKD is strong enough to merit lead chelation testing in patients with CKD who have gout and risk for lead exposure. CaNa_2EDTA is administered (2 doses of 0.5 g in 250 mL 5% dextrose given 12 hours apart), and urine is collected for 3 days because urinary excretion is slower when eGFR is reduced. Normal urinary lead levels are less than 650 $\mu\text{g}/3$ days. x-Ray fluorescence, which provokes the emission of fluorescent photons from the target area, is an alternative method that detects increased bone lead levels, which are also a reflection of cumulative lead exposure. Although x-ray fluorescence measurements allow a rapid, noninvasive estimation of lead in bone, detection equipment is available at only a small number of centers.

Treatment

Treatment involves infusions of CaNa_2EDTA together with removal of the source of lead. The likelihood of a satisfactory response to CaNa_2EDTA is influenced by the degree of interstitial fibrosis that has already occurred.

In industrial and occupational settings, such as in foundry workers and individuals working with lead-based paints and glazes, preventive measures to minimize exposure and low-level absorption are essential. Some studies in children show success with the oral chelating agent succimer (Chemet). Chelation therapy may slow progressive CKD, even in patients with mild lead intoxication.³⁷ However, chelation therapy has not been widely used because of adverse drug effects and concerns about the effects of remobilized lead. It is generally not indicated for adults with blood lead concentrations of less than 45 $\mu\text{g}/\text{dL}$. Because of a lack of clinical trials demonstrating the efficacy and safety of chelation, recommendations for treatment with chelating agents are empiric and may be controversial.

Other Heavy Metal–Induced Nephropathies

Cadmium is a metal with a wide variety of industrial uses, including the manufacture of glass, metal alloys, and electrical equipment. Cadmium is preferentially concentrated in the kidney, principally in the proximal tubule, in the form of a cadmium-metallothionein complex that has a biologic half-life of about 10 years. Cadmium contamination may be an important contributor to the high risk for chronic interstitial nephritis in some agricultural communities in the developing world. A major outbreak of cadmium toxicity occurred in Japan as a result of industrial contamination. The disease was called *itai-itai*, or “ouch-ouch,” because bone pain was the major clinical manifestation. Other manifestations included proximal tubular dysfunction (Fanconi syndrome), hyperphosphaturia, and vitamin D-resistant rickets with osteomalacia. The mechanism by which cadmium elicits chronic inflammation and fibrosis in the kidney is relatively unstudied. The diagnosis is suggested by a history of occupational exposure, increased urinary β_2 -microglobulin, and increased urinary cadmium levels (>7 μg of cadmium per 1 gram of creatinine). Once manifested, kidney injury tends to be progressive, even if exposure is discontinued. Chelation is not effective in humans, and prevention is the only effective treatment.

Arsenic, used as a poison gas in World War I, is present in insecticides, weed killers, wallpaper, and paints. Chronic arsenic toxicity from inorganic arsenic most commonly manifests as sensory and motor neuropathies, distal extremity hyperkeratosis, palmar desquamation, diarrhea and nausea, Aldrich-Mees lines (white bands on the nails), and anemia. In rare cases, it may cause kidney disease, manifested by proximal RTA and chronic interstitial fibrosis. Diagnosis is made by demonstration of an elevated urinary inorganic arsenic level.³⁸

Mercury is found in alloy plants, mirror plants, and some batteries, and mercury intoxication usually occurs as a result of accidental exposure to mercury vapor. Mercury has been shown to induce membranous nephropathy (MN) in experimental animals and has been reported with the use of mercury-containing skin-lightening creams.³⁹ Neither elemental mercury nor the mercurous salt (Hg_2Cl_2) produces sustained renal tubular injury, but mercury dichloride (HgCl_2) may produce acute tubular necrosis and subsequent chronic interstitial nephritis in laboratory animals. However, a report of endemic methyl mercury poisoning in Japan (Minamata disease) revealed a clinical picture dominated by neurologic sequelae; kidney disease in these patients was surprisingly benign, consisting only of tubular proteinuria without changes in serum creatinine.

RADIATION NEPHRITIS

Definition and Epidemiology

Although radiation nephritis was relatively common decades ago, the incidence has decreased considerably because recognition of radiation-induced renal damage has altered protocols for the administration of therapeutic radiation. In general, direct exposure of the kidney to 20 to 30 Gy (1 Gy = 100 rad) over 5 weeks or less will produce radiation nephritis.

Pathology

In general, vascular and glomerular lesions of thrombotic microangiopathy may predominate. The initial target of ionizing radiation within the kidney appears to be endothelial cells, leading to endothelial cell swelling. Electron microscopy (EM) reveals a split appearance of the capillary wall caused by the mesangial interposition and widening of the subendothelial space by a nondescript fluffy material. These features are shared by hemolytic uremic syndrome and thrombotic thrombocytopenic purpura, suggesting a common pathogenic mechanism originating from endothelial injury. However, tubulointerstitial changes are also usually present. Vascular occlusion leads to tubular atrophy, and severe disease is characterized by progressive interstitial fibrosis and the presence of interstitial inflammatory cells.

Clinical Manifestations

Hypertension is commonly observed. Progression to a chronic form of radiation nephritis may occur if resolution of acute radiation nephritis is incomplete. These patients present with proteinuria, progressive CKD, and eventual development of ESKD several years after irradiation in the absence of an acute phase.

Treatment

Prevention is the best approach; risk can be minimized by shielding the kidneys or fractioning the total-body irradiation into several small doses over several days. No specific treatment is available for established radiation nephritis. The general approach is control of hypertension and supportive treatment of CKD.

INTERSTITIAL NEPHRITIS MEDIATED BY IMMUNOLOGIC MECHANISMS

Sjögren Syndrome

Definition and Epidemiology

Sjögren syndrome may be associated with chronic interstitial nephritis. The reported prevalence of kidney involvement in Sjögren syndrome varies from 2% to 67%, principally because of different definitions between studies. Recent analysis of 130 patients with primary Sjögren syndrome in China showed an 80% incidence of biopsy-proven chronic interstitial nephritis.⁴⁰

Pathology

The lesion is characterized histologically by infiltration of lymphocytes and plasma cells in the interstitium with tubular cell injury and, rarely, granuloma formation. This progresses to tubular atrophy and interstitial fibrosis over time. IF reveals granular deposits of immunoglobulin G (IgG) and C3 along the tubular basement membrane (TBM).

Clinical Manifestations

The clinical and biochemical manifestations of interstitial nephritis may be the presenting or only features of Sjögren syndrome. Serum creatinine concentration is generally only mildly elevated in association

with a bland urine sediment and abnormalities in tubular function, including Fanconi syndrome, distal RTA, hypokalemia, and nephrogenic diabetes insipidus. Sjögren syndrome is one of the most common causes of acquired distal (type 1) RTA in adults, and the hypokalemia may be marked, resulting in a clinical presentation of severe weakness. Hypokalemia may occur in the absence of RTA, resulting from salt wasting and secondary hyperaldosteronism.

Treatment

Treatment with corticosteroids at the stage of cellular infiltration is frequently beneficial for protecting kidney function. Although CKD tends to slowly progress over time, ESKD is relatively rare.

Sarcoidosis

Definition and Epidemiology

Histologic evidence of interstitial nephritis with noncaseating granulomas is common in patients with sarcoidosis, but the frequency of clinically significant disease is low.⁴¹ It may manifest as either acute interstitial nephritis or chronic interstitial nephritis.

Pathogenesis and Pathology

Kidney biopsy reveals normal glomeruli; interstitial infiltration, mostly with mononuclear cells; tubular injury; and, with more chronic disease, interstitial fibrosis. Whereas the classic finding is noncaseating granulomas in the interstitium, they are uncommon and nonspecific. An analysis of 18 patients with granulomatous interstitial nephritis showed that in 5 the disorder was associated with sarcoidosis; in 2, tubulointerstitial nephritis and uveitis; in 2, medication; and in 9, the condition was idiopathic.⁴² IF and EM studies typically show no immune deposits.

Clinical Manifestations

Most affected patients have clear evidence of diffuse active sarcoidosis, although some present with an isolated elevation in serum creatinine and only minimal extrarenal manifestations. The urinalysis may be normal or show only sterile pyuria or mild proteinuria.

In addition, hypercalcemia induced by increased production of calcitriol (1,25-dihydroxyvitamin D₃) by activated mononuclear cells (particularly macrophages) in the lung and lymph nodes occasionally results in renal problems (see the previous discussion of hypercalcemic nephropathy).

The serum ACE level is used best not as a primary diagnostic tool but as a marker of disease activity and response to therapy. A normal serum ACE level does not exclude sarcoidosis of the kidney.

Treatment

Corticosteroid therapy tends to improve kidney function, although recovery is often incomplete. Limited data are available concerning the optimal protocol for corticosteroid therapy, but initial dose is usually prednis(ol)one 0.5 to 1 mg/kg/day, which is followed by a slow taper. Rapid tapering of corticosteroids can result in relapse.

Systemic Lupus Erythematosus

Definition and Epidemiology

Interstitial nephritis with immune complexes is defined by granular deposits of immunoglobulins and complement in the TBM, interstitium, or both. Systemic lupus erythematosus is the most common reason for this type of interstitial nephritis (Fig. 65.7), and interstitial involvement is present in half of kidney biopsy specimens in lupus. Tubulitis serves as an independent predictive factor for eGFR decline in lupus nephritis.⁴³ Rarely, tubulointerstitial immune complex disease may be the only manifestation of lupus nephritis.

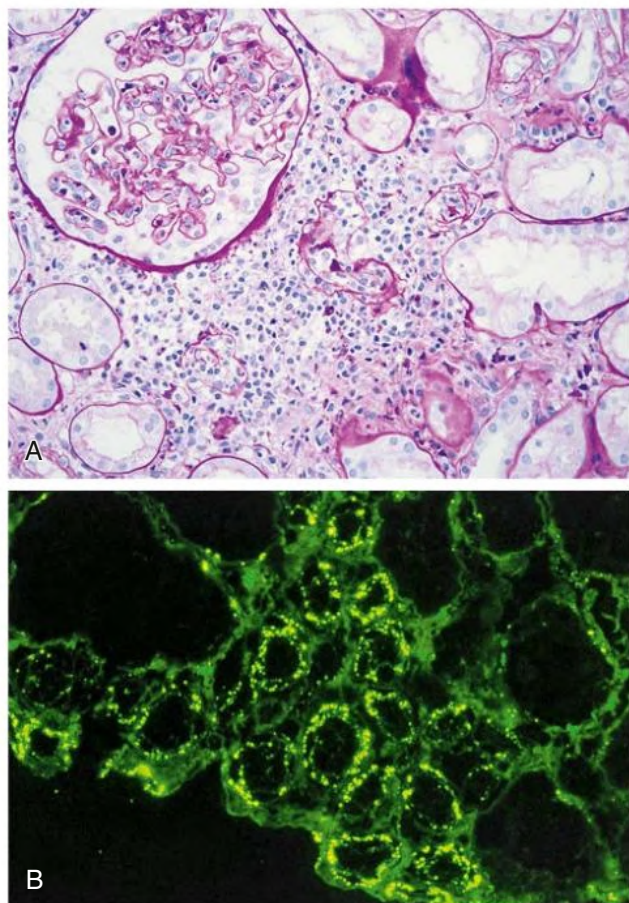


Fig. 65.7 Chronic Interstitial Nephritis in Systemic Lupus Erythematosus. (A) Interstitial nephritis observed in patients with systemic lupus erythematosus. (PAS stain; original magnification $\times 400$.) B, Immunofluorescence study of the same patient revealed deposition of immunoglobulin G in the interstitium, in tubular cells, and along the tubular basement membrane. (Courtesy Drs. Akira Shimizu and Hideki Takano, Nippon Medical School, Tokyo.)

Clinical Manifestations

The presentation may be as acute or chronic interstitial nephritis. The possibility of interstitial involvement (without glomerular disease) is suggested by a rising serum creatinine concentration and a bland urine sediment. Interstitial involvement may be accompanied by signs of tubular dysfunction, such as distal RTA (type 1 or type 4); by isolated hyperkalemia resulting from impaired distal potassium secretion; or by hypokalemia resulting from salt wasting. The potentiating effects of sodium wasting on potassium secretion include an increase in sodium delivery to the potassium secretory site in the collecting tubules and associated volume depletion with subsequent stimulation of aldosterone secretion.

Treatment

Corticosteroid therapy is usually effective in suppressing tubular dysfunction and preserving renal function.

Inflammatory Bowel Disease

Although the most frequent kidney-related complications of Crohn disease are calcium oxalate stones and renal amyloidosis, interstitial nephritis has been reported in patients treated for chronic inflammatory bowel disease. Aminosalicylates (5-aminosalicylic acid, mesalazine, and sulfasalazine) are responsible for most of these cases, but

nephrotoxicity of these agents is very uncommon (mean rate only 0.3% per patient-year).⁴⁴ The median time for development of kidney injury after commencing aminosalicylates is 3 years.⁴⁵ There is no clear relationship between aminosalicylate dose and the risk for nephrotoxicity, suggesting that this is an idiosyncratic response. Some patients have been reported to have biopsy-proven interstitial nephritis before the diagnosis of Crohn disease.

Aminosalicylates should be withdrawn when reduced eGFR develops in a patient with inflammatory bowel disease; if this does not result in a fall in serum creatinine, renal biopsy should be considered. Only one-third of cases recover completely after drug withdrawal.⁴⁵ Corticosteroids are recommended when kidney function does not respond to drug withdrawal.

Immunoglobulin G4-Related Kidney Disease

Definition and Epidemiology

On the basis of histologic and immunohistochemical examination of various organs of patients with autoimmune pancreatitis, a novel clinicopathologic entity of immunoglobulin (Ig) G4-related sclerosing disease has been proposed.⁴⁶ This systemic disease is characterized by extensive IgG4-positive plasma cells and T-lymphocyte infiltration of various organs (Fig. 65.8). Initial reports of IgG4-related sclerosing disease came from Japan, but it has now been described in Europe and the United States and is considered to be a worldwide entity.

Pathogenesis

Whether IgG4 is pathogenic or is a “bystander” remains unknown. IgG4 does not activate the classical complement pathway effectively. However, immune complex formation may play a pathogenic role, raising the possibility of complement fixation via the lectin pathway or activation of the classical pathway of complement by some unknown mechanism.

Pathology

The most common pattern of involvement by IgG4-related kidney disease is tubulointerstitial nephritis with dense infiltration of IgG4-positive mononuclear cells. Distinctive features of the tubulointerstitial nephritis include: (1) well-demarcated borders between involved and uninvolved areas; (2) involvement of the cortex and deep medulla, often extending beyond the renal capsule; (3) interstitial inflammatory cells made up predominantly of plasma cells and lymphocytes, with a high prevalence of IgG4-positive cells often admixed with fibrosis; (4) peculiar features of interstitial fibrosis called *storiform fibrosis* or *bird's-eye* pattern; and (5) deposits visible by light microscopy and IF in the TBM, Bowman's capsule, and interstitium.⁴⁷ Recent pathologic analysis revealed perivascular inflammation or fibrosis of medium- and small-sized vessels as a newly identified pathologic feature of IgG4-related kidney disease and proposed the term *interstitial fibrosclerosis* because storiform fibrosis contains mainly nonfibrillar collagens.⁴⁸ IF shows granular TBM staining for IgG accompanied by C3 of lesser staining intensity. MN also has been described as a manifestation of IgG4-related kidney disease.⁴⁹

Clinical Manifestations

Clinical manifestations are observed in various organs, with presentations including sclerosing cholangitis, cholecystitis, sialadenitis, and retroperitoneal fibrosis. IgG4-related tubulointerstitial nephritis can be mass forming, similar to IgG4-related inflammatory lesions in other organs. Of patients with IgG4-related disease, 80% are reported to have had radiographic abnormalities of the kidneys: bilateral and multiple small low-attenuation lesions, a mass, or bilateral kidney enlargement.⁵⁰

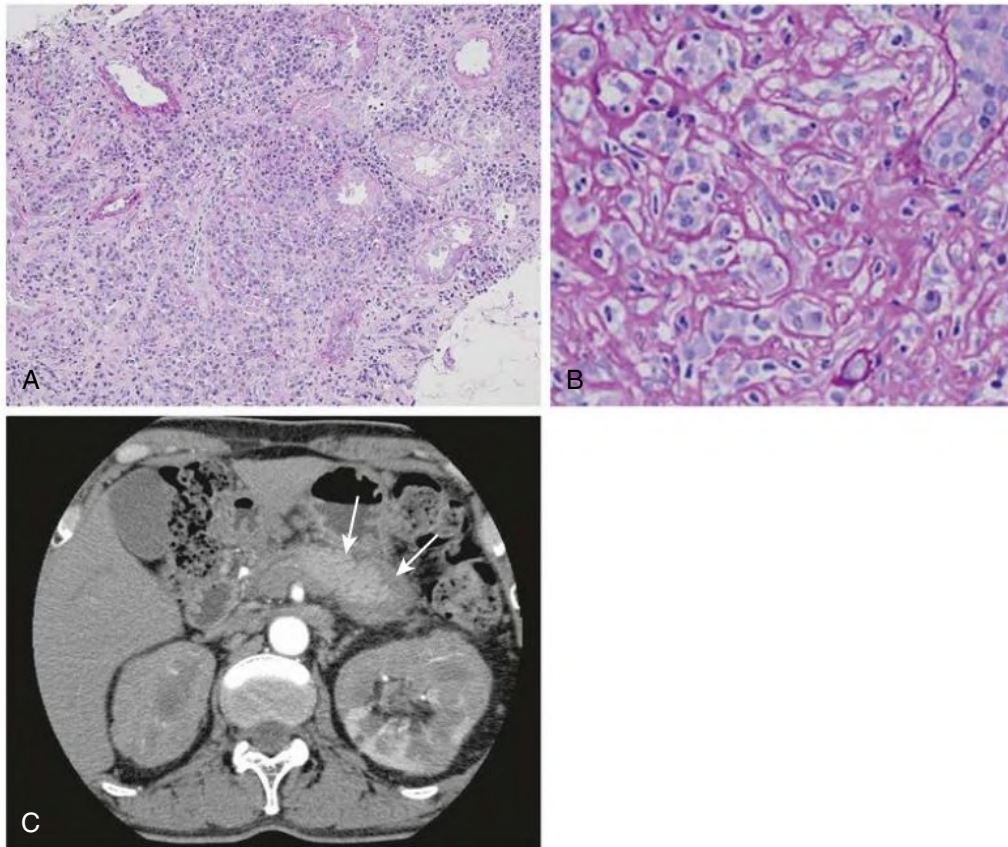


Fig. 65.8 Chronic Interstitial Nephritis in Immunoglobulin (Ig) G4-Related Sclerosing Disease. (A) Interstitial nephritis with numerous mononuclear cell infiltrates observed in a patient with autoimmune pancreatitis. Many of these cells are positive for IgG4 in typical cases. (PAS stain; original magnification $\times 200$.) (B) Small nests of swollen plasma cells or individual plasma cells are encircled by collagenous tissue. The features resemble a maple wood grain pattern called “bird’s eye.” Plasma cells look like birds’ eyes in the wood, whereas fibrotic tissues correspond to branches of the wood. (C) Computed tomography scan of the patient revealed pancreatic swelling (arrows). (A and C, Courtesy Dr. Hiroshi Nishi, University of Tokyo, Tokyo. B, From Joss N, Morris S, Young B, Geddes C. Granulomatous interstitial nephritis. *Clin J Am Soc Nephrol.* 2007;2:222–230.)

Although most IgG4-related sclerosing disease is associated with autoimmune pancreatitis, cases without pancreatic involvement have been described. The disease occurs predominantly in older males. Serum IgG4 levels are raised, and IgG4-positive cells are found in the interstitium. According to a recent meta-analysis, a cut-off value of serum IgG4 ranging from 135 to 144 mg/dL confers a sensitivity of 87% and a specificity of 83%.⁵¹ However, the finding of an elevated serum IgG4 is not specific because this may occur in 5% of the normal population. The patients often show hypocomplementemia and eosinophilia. Recently proposed diagnostic criteria are shown in [Table 65.3](#).

Treatment

The response to corticosteroids is generally favorable, but reappearance of clinical manifestations is common. The protocol has not been established, but a starting dose of approximately 40 mg/day may be prudent. There is no correlation between histologic pattern and response to therapy, and even patients with extensive fibrosis on biopsy have shown a response to corticosteroid therapy. Rituximab may be an alternative treatment for IgG4-related kidney disease. In a recent open-label pilot trial of 30 patients, two doses of rituximab (1000 mg each) achieved the primary outcome composed of improvement of disease

activity and damage, no disease flares before month 6, and no steroid use between months 2 and 6, in 77% of participants.⁵²

Other Forms of Immune-Mediated Interstitial Nephritis

Primary anti-TBM nephritis is an extremely rare form of interstitial nephritis that usually is acute and characterized by linear deposits of immunoglobulins, commonly IgG, and complement in the TBM, together with tubular interstitial inflammation and anti-TBM antibodies in the serum. Anti-TBM antibodies, usually IgG, may be found in 50% to 70% of patients with anti-glomerular basement membrane nephritis and occasionally in patients with MN, systemic lupus, IgA nephropathy, minimal change disease, and malignant hypertension. The antibodies are directed against a protein that is found in the proximal TBM and regulates tubulogenesis.

OBSTRUCTIVE UROPATHY

Complete or partial urinary tract obstruction is accompanied by pathologic changes in both the tubulointerstitium and glomeruli consisting of interstitial fibrosis, tubular atrophy, and occasionally focal glomerular sclerosis. Details are discussed in [Chapter 61](#).

TABLE 65.3 Diagnostic Criteria for IgG4-Related Tubulointerstitial Nephritis

Histology	Plasma cell–rich tubulointerstitial nephritis with >10 IgG4-positive plasma cells per high-power field in the most concentrated field ^a Tubular basement membrane immune complex deposits by immunofluorescence, immunohistochemistry, and/or electron microscopy ^b
Imaging	Small peripheral low-attenuation cortical nodules, round or wedge-shaped lesions, or diffuse patchy involvement Diffuse marked enlargement of kidneys
Serology	Elevated serum IgG4 or total IgG level
Other organ involvement	Includes autoimmune pancreatitis, sclerosing cholangitis, inflammatory masses in any organ, sialadenitis, inflammatory aortic aneurysm, lung involvement, retroperitoneal fibrosis

Diagnostic criteria for IgG4-related TIN. Diagnosis of IgG4 TIN requires the histologic feature of plasma cell–rich TIN with increased IgG4-positive plasma cells and at least one other feature from the categories of imaging, serology, or other organ involvement.

^aMandatory criterion.

^bSupportive criterion, present in >80% of cases.

IgG, Immunoglobulin G; TIN, tubulointerstitial nephritis.

From Heap GA, So K, Weedon M, et al. Clinical features and HLA association of 5-aminosalicylate (5-ASA)-induced nephrotoxicity in inflammatory bowel disease. *J Crohns Colitis*. 2016;10:149–158.

VASCULAR DISEASES

Ischemia resulting from intrarenal vascular involvement causes tubular atrophy, interstitial fibrosis, and cellular infiltration. This is further discussed in [Chapter 43](#). Chronic ischemia in the tubulointerstitial compartment also plays a crucial role in the progression of a variety of glomerular and tubulointerstitial diseases.²

INFECTION-ASSOCIATED CHRONIC INTERSTITIAL NEPHRITIS

Although a variety of bacterial and viral infections can be associated with acute interstitial nephritis (see [Chapter 64](#)), chronic interstitial nephritis secondary to infectious agents appears to be rare. Insidious *Mycobacterium tuberculosis* infection can cause chronic granulomatous tubulointerstitial nephritis.⁵³ Chronic bacterial infections can result in xanthogranulomatous pyelonephritis or renal malacoplakia (see [Chapter 53](#)). In mice, an atypical virus termed mouse kidney parvovirus is capable of driving a stepwise progression of pathology ranging from sporadic tubular inclusions to tubular degeneration and interstitial fibrosis.⁵⁴

SELF-ASSESSMENT QUESTIONS

1. A 29-year-old woman was admitted to the hospital because of anorexia, binge eating, and self-induced vomiting. On admission, her body mass index was 12.2 and she presented with nocturia and polyuria. The serum potassium level was 2.8 mEq/L. Which one of the following do you expect as the pathologic change of the kidney biopsy specimen of this patient?
 - A. Calcific deposits in the interstitium
 - B. Vacuolation of the kidney tubules
 - C. Noncaseating granulomas
 - D. Infiltration of IgG4-positive mononuclear cells
 - E. Acellular interstitial nephritis with intranuclear inclusion bodies in the proximal tubules
2. IgG4-related tubulointerstitial nephritis is an emerging clinico-pathologic entity that is a part of the systemic disease characterized by infiltration of IgG4-positive plasma cells in various organs. Which one of the following is *most* likely to be associated with this disease?
 - A. Uroepithelial carcinoma
 - B. Inflammatory bowel disease
 - C. Cerebral aneurysm
 - D. Autoimmune pancreatitis
 - E. Angiokeratoma
3. A 58-year-old woman with a 30-year history of bipolar affective illness was admitted to hospital because of thirst, polyuria, and weight loss. Lithium had been prescribed for the preceding 18 years. On admission, she presented with hypernatremic dehydration. Which one of the following represents the *best* next step to diagnose the characteristic morphologic changes in the kidney?
 - A. Non-contrast-enhanced MRI
 - B. Blood oxygen level-dependent MRI
 - C. Non-contrast-enhanced computed tomography
 - D. Scintigraphy
 - E. Intravenous urography

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